## TOPIC 1 Linear Equations and Inequalities

Name _____ Date _____

**Reminders**
**Standard → slope-intercept:** subtract the x-term, then divide every term by the number with y.
**Point-slope → slope-intercept:** distribute, then add or subtract to get y alone.
**Intercepts:** x-intercept: put 0 in for y. y-intercept: put 0 in for x.
**Solve a formula for a letter:** undo everything around that letter, doing the same to both sides.

**1.** What value of $x$ makes $2(x + 3) = x - 4$ true?

**Tip:** distribute, then get [[x]] on one side

**A.** $-10$

**B.** $-7$

**C.** 2

$2x + 6 = x - 4$, so $x = -10$.

**2.** How many solutions does $3x + 1 = 3x - 5$ have?

**Tip:** same [[x]]-term + different numbers → no solution

**F.** One solution, $x = 0$

**G.** No solution

**H.** Infinite solutions

Same $x$-term, different constants: $1 \neq -5$, never true.

**3.** What are the slope and $y$-intercept of $y = -2x + 7$?

> **Tip:** [[y = mx + b]]: slope [[m]], [[y]]-intercept [[(0, b)]]

**A.** slope 7, (0, −2)

**B.** slope 2, (0, 7)

**C. slope −2, (0, 7)**

In $y = mx + b$, $m = -2$ and $b = 7$.

**4.** What is the $x$-intercept of $y = 2x - 6$?

> **Tip:** [[x]]-intercept: put 0 in for [[y]]

**F.** (−6, 0)

**G.** (0, −6)

**H. (3, 0)**

Set $y = 0$: $0 = 2x - 6$, $x = 3$.